

A second order unstructured finite volume method for incompressible Navier-Stokes equations using face based constrained least-squares procedure

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Abstract

This paper discusses the use of face based constrained least-squares procedure for solving incompressible Navier-Stokes equations. Although reconstruction is performed on faces of the mesh, the underlying finite volume method is still cell centered. This arrangement of reconstruction is particularly advantageous in implementation

1. Methodology

1.1. Constrained Least Squares

The constrained least squares is [1] used for reconstruction [3] to solve incompressible Navier-Stokes [2]

1.2. Reconstruction for interior faces

The constrained least-squares system for reconstruction on an interior face of the mesh is

$$\begin{bmatrix} 2(N_c - 2) & 2 \sum \Delta x_i & 2 \sum \Delta y_i & 1 & 1 \\ 2 \sum \Delta x_i & 2 \sum \Delta x_i^2 & 2 \sum \Delta x_i \Delta y_i & \Delta x_1 & \Delta x_2 \\ 2 \sum \Delta y_i & 2 \sum \Delta x_i \Delta y_i & 2 \sum \Delta y_i^2 & \Delta y_1 & \Delta y_2 \\ 1 & \Delta x_1 & \Delta y_1 & 0 & 0 \\ 1 & \Delta x_2 & \Delta y_2 & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_x \\ \phi_y \\ \hat{z}_1 \\ \hat{z}_2 \end{bmatrix} = \begin{bmatrix} 2 \sum \phi_i \\ 2 \sum \Delta x_i \phi_i \\ 2 \sum \Delta y_i \phi_i \\ \phi_1 \\ \phi_2 \end{bmatrix} \quad (1)$$

1.3. Reconstruction for faces on Dirichlet boundaries

$$\begin{bmatrix} 2(N_c - 2) & 2 \sum \Delta x_i & 2 \sum \Delta y_i & 1 & 1 \\ 2 \sum \Delta x_i & 2 \sum \Delta x_i^2 & 2 \sum \Delta x_i \Delta y_i & \Delta x_1 & 0 \\ 2 \sum \Delta y_i & 2 \sum \Delta x_i \Delta y_i & 2 \sum \Delta y_i^2 & \Delta y_1 & 0 \\ 1 & \Delta x_1 & \Delta y_1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_x \\ \phi_y \\ \hat{z}_1 \\ \hat{z}_2 \end{bmatrix} = \begin{bmatrix} 2 \sum \phi_i \\ 2 \sum \Delta x_i \phi_i \\ 2 \sum \Delta y_i \phi_i \\ \phi_1 \\ \phi_b \end{bmatrix} \quad (2)$$

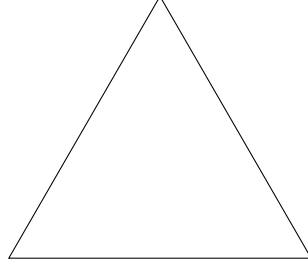


Figure 1: Arrangement of variables in a cell

1.4. Reconstruction for faces on Neumann boundaries

$$\begin{bmatrix} 2(N_c - 2) & 2 \sum \Delta x_i & 2 \sum \Delta y_i & 1 & 0 \\ 2 \sum \Delta x_i & 2 \sum \Delta x_i^2 & 2 \sum \Delta x_i \Delta y_i & \Delta x_1 & n_x \\ 2 \sum \Delta y_i & 2 \sum \Delta x_i \Delta y_i & 2 \sum \Delta y_i^2 & \Delta y_1 & n_y \\ 1 & \Delta x_1 & \Delta y_1 & 0 & 0 \\ 0 & n_x & n_y & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_x \\ \phi_y \\ \hat{z}_1 \\ \hat{z}_2 \end{bmatrix} = \begin{bmatrix} 2 \sum \phi_i \\ 2 \sum \Delta x_i \phi_i \\ 2 \sum \Delta y_i \phi_i \\ \phi_1 \\ \phi_b \end{bmatrix} \quad (3)$$

All summations in the matrix above are from $i = 3$ to N_c

References

- [1] S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press, 2018.
- [2] Dongjoo Kim and Haecheon Choi. A second-order time-accurate finite volume method for unsteady incompressible flow on hybrid unstructured grids. *Journal of Computational Physics*, 162(2):411–428, 2000.
- [3] Emre Sozer, Christoph Brehm, and Cetin C. Kiris. *Gradient Calculation Methods on Arbitrary Polyhedral Unstructured Meshes for Cell-Centered CFD Solvers*.